

A Comparison of Beta-Carotene Levels in Patients Overtime by Dosage Group

By: Cameron Fong

Contents

1	Abstract.....	3
2	Introduction.....	3
3	Methods.....	3
3.1	Sampling Scheme.....	3
3.2	Available Covariates.....	4
3.3	Statistical Methods.....	4
4	Results.....	7
4.1	Features of the data.....	7
4.2	Association between dosage of beta carotene and serum beta carotene levels.....	8
4.3	Relationship between dosage of beta carotene and beta carotene levels over time.....	8
4.4	Relationship between dosage of beta carotene and rate back to baseline.....	9
4.5	Relationship of age, male, bmi, and cholesterol with interaction of month and dose.....	9
4.6	Beta carotene level Prediction Modeling for months 10-12.....	10
5	Discussion.....	10
6	Appendix.....	11

1 Abstract

This study examines whether beta-carotene supplementation dose-dependently impacts serum beta-carotene levels over time and how quickly levels return to baseline after supplementation. Using a 2 sample t-test and 1 way anova test, we concluded that taking the supplementation increases time averaged beta carotene levels by approximately 645 to 1300 beta carotene levels (BCL) with a p-value of 6.9×10^{-5} , and that this effect is dose dependent (higher dosage led to higher time averaged BCL). Primarily using linear mixed effects models and interaction analyses of longitudinal supplementation data, we found that although the 60 dose group elicited the strongest initial response (increase of 554.60 BCL, with a p-value of 0.006 over baseline supplementation of 15), higher doses showed no additional difference in BCL overtime. Post-supplementation, the 45 dose group declined most sharply (-222 BCL per month at 12 months compared to dose 15, with a p-value of 0.011), while lower doses such as dose 15 and 30 had approximately the same rate of decrease to baseline. Gender significantly modified treatment effects over time with a p-value of 0.002, but age, BMI, and cholesterol did not. Key limitations include wide confidence intervals (e.g., ± 500 -700 BCL for individual predictions) and short follow-up. These findings highlight the importance of dose selection and monitoring, particularly for intermediate doses such as dose 45, which show rapid decreasing rates after stopping supplementation. Future studies should investigate metabolic mechanisms underlying these non-linear dose-response patterns.

2 Introduction

Beta-carotene, an optimal, naturally occurring, provitamin A [1], has been proposed as a potential preventive measure against cancer. However, before large-scale clinical trials can assess its efficacy in reducing cancer risk, it is essential to understand its pharmacokinetics - the four stages of absorption, distribution, metabolism, and excretion [2] of beta-carotene at different doses. Additionally, since beta-carotene and vitamin E are both fat-soluble compounds, there is a possibility that beta-carotene supplementation could influence vitamin E levels in the blood, which may have broader implications for nutrient balance and health. Linear mixed effect model is implemented to show the association between beta carotene levels within subjects and the dose level of beta carotene administered to subjects alongside other covariates.

The primary objectives of this analysis are to answer the following questions of interest:

1. Is supplementation of beta-carotene associated with an increased time-average of serum beta-carotene levels? If so, is the effect of supplementation dose-dependent?
2. Does beta-carotene supplementation impact serum beta-carotene levels over time, and if so, is the impact dose-dependent?
3. If there are changes to serum beta-carotene levels due to supplementation, are there differences by dose in the rate at which patients return to baseline after supplementation is stopped?
4. Quantify whether the effect of beta-carotene supplementation on serum beta-carotene levels over time differ by age, gender, body mass index (BMI), or cholesterol.
5. For two randomly sampled patients at each dose level, provide predictions of their serum beta-carotene levels if they were to have stayed on treatment for an additional 3 months.

By addressing these questions, this study aims to provide critical insights that will inform future clinical trials on beta-carotene's potential role in cancer prevention.

3 Methods

3.1 Sampling Scheme

The dataset we will analyze contains experimental data from $N = 46$ volunteers who were randomly assigned to one of five doses of beta-carotene supplementation (0, 15, 30, 45, or 60 dose) in a double-blind Phase II clinical trial. Serum beta-carotene and vitamin E levels were measured at multiple time points before, during, and after treatment. Additional covariates include subject age, sex, BMI, and baseline serum cholesterol levels.

An ideal pharmacokinetic study would involve randomized dose assignment with sufficient follow-up to assess drug absorption, metabolism, and clearance. Our study aligns with this in some key ways as well as the design itself including strengths such as participants being randomly assigned to dose groups to minimize selection bias, the study including a pre-treatment phase (months 0–3), an active treatment phase (months 4–9), and a post-treatment washout phase (months 10–15) thus allowing for assessment of drug buildup and clearance, and serum levels being tracked longitudinally which enables analysis of time-dependent effects.

The study does also have limitations affecting the interpretation of results including the sample size being only 46 participants where statistical power to detect subtle dose-response relationships or interactions (e.g., with BMI or sex) may be limited, the washout period extended only to month 15 meaning longer follow-up might be necessary to confirm full clearance, unmeasured factors (example being genetic differences in metabolism) could still influence serum levels hence potential confounding.

In addition, due to the controlled experimental design, conclusions about beta-carotene pharmacokinetics (e.g., dose-response relationships, time to peak concentration, and clearance rates) can reasonably be extended to similar healthy adult populations. For vitamin E levels, the study can assess whether beta-carotene supplementation affects serum concentrations, but causal inference is limited by potential unmeasured confounders.

3.2 Available Covariates

In terms of pharmacokinetics in general, factors that affect the rate of pharmacokinetics in a person can include the properties within the beta carotene (including dosage amount), age, gender, physiology like obesity or dehydration, metabolism of a person, body composition (total body water, muscle mass, body fat), etc. [3][4] In terms of the available data within this data set, we have:

1. Id: patient identifier
2. Month: the study month of a particular patient split up into the untreated phase (months 0 to 3), treatment phase (months 4 to 9) where subjects were given their respective dose, and the after supplementation phase (months 10-15). This will be used to see the rate of beta carotene levels within a subjects' body over time
3. Bcarot: serum beta carotene level in a subject in terms of ug/mL. This is the response variable of interest that will help in answering the questions of interest.
4. Vite: serum vitamin E within a subject measured in ug/mL
5. Dose: dose of beta carotene supplementation given within the months of 4 to 9 and is factorized by dose amounts of 0, 15, 30, 45, and 60
6. Age: the age of the subject in years at the baseline or start of the trial at month 0
7. Male: an indicator with value 1 if the subject is male or value 0 if the subject is female
8. Bmi: the subject's body mass index measured in kg / height in m² at baseline of the trial at month 0
9. Chol: subject serum cholesterol at baseline or month 0

With the known variables, month (time), vite, dose, age, gender (male), bmi, and chol are all known to affect b carotene levels within individual subjects. Since subjects were randomly assigned to one of the 5 beta carotene doses (0,15,30,45, and 60), confounding variables should be unlikely but regardless will be included to help with robustness.

The data also notably has some missing values with bcarot and vite, both missing with the same frequency of 6 times for any instance of a month for any subject with subject 12 having the most missing data with 2 months of no documented bcarot or vite in months 0 and 11. Despite this, the data appears to be missing at random with no noticeable patterns of missingness within the other variables of chol, bmi, gender, age, dose group, or particular month. In addition, we test our questions of interest using linear mixed models which by design are able to handle missing data under MAR (missing at random) therefore we can retain these missing cases as well as deal with overdispersion via flexible variance modeling.

3.3 Statistical Methods

We build 5 distinct models in order to answer the five questions of interest shown in the introduction.

To answer question 1, we calculate each subject's change in serum beta-carotene by subtracting their baseline average (months 0-3) from their treatment-phase average (months 4-9), creating a derived outcome that accounts for individual starting levels. Using Welch's two-sample t-test (which accommodates unequal group sizes and variances), we

compared these changes between the non-supplemented group (0 dose) and supplemented groups (combined 15-60 dose groups). For significant findings, we proceeded to test for dose-dependency by grouping subjects into their specific dosage categories (15, 30, 45, and 60 doses) and conducting a one-way ANOVA on the same change scores.

To answer question 2 part 1, a linear mixed model was used with fixed effects month and treatment which is a factor in which treatment = 0 if a subject has dose 0 and treatment = 1 if a subject has any dose 15,30,45, or 60, as well as the inclusion of interaction term treatment and month (our covariate of interest), the random effect intercept to account for baseline differences between individuals in their serum β -carotene levels, and the baseline covariates age, male, bmi, and chol to account for confounding. This model was subsetted to months 4 to 9 as we are only interested in the months at which the beta carotene is supplemented shown below where Y_{ij} is the outcome variable beta carotene.

$$\begin{aligned} \text{bcarot}_{ij} = & \beta_0 + \beta_1 \text{treatment}_i \\ & + \beta_2 \text{month}_{ij} + \beta_3 \text{age}_i + \beta_4 \text{male}_i \\ & + \beta_5 \text{bmi}_i + \beta_6 \text{chol}_i \\ & + \beta_7 \text{treatment}_i \times \text{month}_{ij} \\ & + b_{0i} + \varepsilon_{ij} \end{aligned}$$

Model Fit for Question 2 Part 1 of Interest

If we conclude beta-carotene supplementation impacts serum beta-carotene levels over time, we consider the doses individually to see if they are an effect modifier, or if different doses lead to different rates of beta carotene levels over time. To do this, we fit the same model as part 1 but replace treatment with the original dose covariate shown below:

$$\begin{aligned} \text{bcarot}_{ij} = & \beta_0 + \beta_1 \mathbb{I}(\text{dose}_i = 30) + \beta_2 \mathbb{I}(\text{dose}_i = 45) + \beta_3 \mathbb{I}(\text{dose}_i = 60) \\ & + \beta_4 \text{month}_{ij} + \beta_5 \text{age}_i + \beta_6 \text{male}_i \\ & + \beta_7 \text{bmi}_i + \beta_8 \text{chol}_i \\ & + \beta_9 \mathbb{I}(\text{dose}_i = 30) \times \text{month}_{ij} \\ & + \beta_{10} \mathbb{I}(\text{dose}_i = 45) \times \text{month}_{ij} \\ & + \beta_{11} \mathbb{I}(\text{dose}_i = 60) \times \text{month}_{ij} \\ & + b_{0i} + \varepsilon_{ij} \end{aligned}$$

Model Fit for Question 2 Part 2 of Interest

To answer question 3, our sample of interest is those who have taken a supplementation of beta carotene as per the question, thus subjects who were administered the placebo are not used to answer the question. We also only use months 10-15 as we are only interested in the rate of beta carotene levels after the supplementation phase has ended. As such, we center the data starting at the point of supplementation ending or month 10 in order for interpretations, notably for the intercept, to make sense in the context of the question which we will call month_centered. In addition, a linear spline with knot at month 12 is included in the model to account for longitudinal changes in months. Like that of question 2, we also use a linear mixed effects model with fixed effects dose, month_centered, month_sp4 (spline at month 12), age, male, bmi, and chol with interaction terms between dose with month_centered and dose with month_sp4 alongside random effects of intercept and month_centered to account for individual variability within baseline level and linear trend of beta carotene. Below is the model to fit for question 3:

$$\begin{aligned} \text{bcarot}_{ij} = & \beta_0 + \beta_1 \mathbb{I}(\text{dose}_i = 30) + \beta_2 \mathbb{I}(\text{dose}_i = 45) + \beta_3 \mathbb{I}(\text{dose}_i = 60) \\ & + \beta_4 \text{month_centered}_{ij} + \beta_5 \text{month_sp4}_{ij} \\ & + \beta_6 \text{age}_i + \beta_7 \text{male}_i + \beta_8 \text{bmi}_i + \beta_9 \text{chol}_i \\ & + \beta_{10} \mathbb{I}(\text{dose}_i = 30) \times \text{month_centered}_{ij} \\ & + \beta_{11} \mathbb{I}(\text{dose}_i = 45) \times \text{month_centered}_{ij} \\ & + \beta_{12} \mathbb{I}(\text{dose}_i = 60) \times \text{month_centered}_{ij} \\ & + \beta_{13} \mathbb{I}(\text{dose}_i = 30) \times \text{month_sp4}_{ij} \\ & + \beta_{14} \mathbb{I}(\text{dose}_i = 45) \times \text{month_sp4}_{ij} \\ & + \beta_{15} \mathbb{I}(\text{dose}_i = 60) \times \text{month_sp4}_{ij} \\ & + b_{0i} + b_{1i} \text{month_centered}_{ij} + \varepsilon_{ij} \end{aligned}$$

Model Fit for Question 3 of Interest

To answer question 4, beta carotene supplementation only affects serum beta carotene levels once the supplementation is administered, thus months 4-15 are of interest and like question 3, we repurpose month_centered to have month 4 be the start instead of month 0. We also implement 2 linear splines month_sp3 and month_sp4 for months 10 and 12 respectively to accommodate the sudden spike and subsequent drop in beta carotene levels after the supplementation phase is finished. The treatment variable from question 2 is also used in place of dose as we want to see each baseline covariates' impact on the interaction of supplementation and time (month) in general rather than by doses independently. To examine how age, sex, BMI, and cholesterol modify the treatment effect over time, we constructed four separate linear mixed-effects models - one for each baseline covariate. Each model incorporated fixed effects for all baseline covariates (age, male, BMI, and cholesterol) to control for potential confounding. We modeled beta carotene trends using both linear (month_centered) and spline terms (month_sp3, month_sp4) to capture potential changes in β -carotene levels across the study period. We also included three-way interactions between the focal covariate (example being age), treatment, and each time term to test whether: (1) the treatment effect varied linearly over time depending on the covariate (via month_centered interactions), and (2) the covariate produced distinct response patterns at spline-defined time points. All models included random intercepts and month_centered slopes to account for individual variability, ensuring proper estimation of our fixed effects of interest. This approach allowed us to rigorously assess whether each baseline characteristic modified the treatment's effect over time while maintaining appropriate control for other factors. An example of age's linear mixed model is shown below where month_sp3 is $month_{ij} - 3$, month_sp4 is $month_{ij} - 4$, $treatment_i = 1$ for subjects with doses 15,30,45, and 60 and = 0 when dose is 0.

$$\begin{aligned} bcarot_{ij} = & \beta_0 + \beta_1 age_i + \beta_2 treatment_i + \beta_3 month_centered_{ij} \\ & + \beta_4 month_sp3_{ij} + \beta_5 month_sp4_{ij} \\ & + \beta_6 male_i + \beta_7 bmi_i + \beta_8 chol_i \\ & + \beta_9 age_i treatment_i + \beta_{10} age_i month_centered_{ij} \\ & + \beta_{11} treatment_i month_centered_{ij} \\ & + \beta_{12} age_i month_sp3_{ij} + \beta_{13} age_i month_sp4_{ij} \\ & + \beta_{14} treatment_i month_sp3_{ij} + \beta_{15} treatment_i month_sp4_{ij} \\ & + \beta_{16} age_i treatment_i month_centered_{ij} \\ & + \beta_{17} age_i treatment_i month_sp3_{ij} \\ & + \beta_{18} age_i treatment_i month_sp4_{ij} \\ & + \beta_{0i} + \beta_{1i} month_centered_{ij} + \varepsilon_{ij} \end{aligned}$$

Model Fit for Question 4 of Interest

3-way anova tests are conducted on each baseline covariate linear mixed effects model with their respective reduced model not including baseline interaction with either treatment nor time term.

To answer question 5, the time variable (month) was centered at the beginning of the observation period (month 4) to ensure the intercept represents baseline beta-carotene levels at the start of our analysis window (months 4-9). The sample of interest is with subjects that had an actual beta carotene supplementation and not placebo (dose = 0) thus we only consider prediction models of months 10-12 for subjects with doses 15,30,45,60. The linear mixed model used for the question incorporates several key components including fixed effects for dose, centered month (month_centered), and baseline covariates (age, sex, BMI, and cholesterol) were included to account for potential confounding factors. The interaction term between dose and month_centered (dose*month_centered) allows us to incorporate the effect of supplementation over time across different dose levels. Random intercepts and slopes for month_centered by participant (1 + month_centered | id) account for individual variability in both baseline beta-carotene levels and the rate of change over time. This modeling approach enables us to estimate population-average trajectories of beta-carotene levels while accounting for between-subject variability and control important demographic and health factors that may influence beta-carotene metabolism. The centered month variable ensures that the intercept represents a meaningful time point in our study (month 4), while the random effects structure acknowledges that participants may start at different baseline levels and respond to treatment at different rates. The model for this question is shown below:

$$\begin{aligned}
\text{bcarot}_{ij} = & \beta_0 + \beta_1 \mathbb{I}(\text{dose}_i = 30) + \beta_2 \mathbb{I}(\text{dose}_i = 45) + \beta_3 \mathbb{I}(\text{dose}_i = 60) \\
& + \beta_4 \text{month_centered}_{ij} + \beta_5 \text{age}_i + \beta_6 \text{male}_i \\
& + \beta_7 \text{bmi}_i + \beta_8 \text{chol}_i \\
& + \beta_9 \mathbb{I}(\text{dose}_i = 30) \times \text{month_centered}_{ij} \\
& + \beta_{10} \mathbb{I}(\text{dose}_i = 45) \times \text{month_centered}_{ij} \\
& + \beta_{11} \mathbb{I}(\text{dose}_i = 60) \times \text{month_centered}_{ij} \\
& + b_{0i} + b_{1i} \text{month_centered}_{ij} + \varepsilon_{ij}
\end{aligned}$$

Model Fit for Question 5 of Interest

4 Results

4.1 Features of the data

On top of the missing data explained in the available covariates, other modifications were made to the original data set including factoring male (gender) and dose variables as they are explicitly defined categories and are not continuous data or numeric as first received in the initial data set. In addition, subject id 26 involves an error with its dosage value in which all months 1-15 for subject 26 has dosage set to 0 while for month 0 dosage was mistakenly set to 30, thus is fixed for consistency.

Once all the adjustments have been made to the initial data set, table 1 shows the numerical summary of the relevant covariates stratified by dose type. The baseline characteristics of participants across the five dose groups (0, 15, 30, 45, and 60) were well-balanced, with no statistically significant differences observed in age, sex distribution, vitamin E levels, BMI, or cholesterol levels (all p-values > 0.05), indicating successful randomization. Mean age was consistent across groups, ranging from 55.88 to 57.44 years, while sex distribution showed relatively equal proportions of males and females in each group (44-67% male). Clinical measures, including vitamin E (mean 7.61-8.20 mg/dL), BMI (mean 24.94-26.18 kg/m²), and cholesterol (mean 213.31-238.05 mg/dL), demonstrated similar distributions across dose levels. Notably, the 60 dose group had the highest mean cholesterol level (238.05 ± 38.88 mg/dL), though this difference was not statistically significant (p-value = 0.468). The variable "board" showed substantial variability within groups (SDs 87-130 units) but no significant between-group differences (p-value = 0.767), warranting closer examination for potential data entry errors. While measured covariates were balanced, unrecorded factors such as dietary habits, smoking status, or genetic predispositions could still influence outcomes and should be considered as potential confounders. The absence of extreme BMI values (all within clinically plausible ranges) suggests good data quality for this metric, though individual outliers should still be reviewed. These findings support the validity of between-group comparisons in subsequent analyses, as the randomization process effectively balanced known baseline characteristics. Plot of the mean b carotene levels for subjects of each of the 5 doses can be found in the appendix.

	level	0 Dose Group	15 Dose Group	30 Dose Group	45 Dose Group	60 Dose Group	p	test
n		9	10	9	8	10		
age (mean (SD))		56.11 (4.04)	56.30 (4.64)	57.44 (4.25)	55.88 (3.14)	56.50 (5.21)	0.953	
male (%)	Female	4 (44.4)	5 (50.0)	6 (66.7)	4 (50.0)	5 (50.0)	0.906	
	Male	5 (55.6)	5 (50.0)	3 (33.3)	4 (50.0)	5 (50.0)		
vite (mean (SD))		8.08 (1.38)	7.61 (1.18)	7.83 (1.64)	7.99 (0.65)	8.20 (1.24)	0.877	
bmi (mean (SD))		26.18 (3.59)	25.69 (3.58)	25.83 (2.66)	25.35 (3.32)	24.94 (2.43)	0.931	
chol (mean (SD))		216.00 (27.17)	223.00 (29.72)	214.44 (35.28)	213.31 (33.54)	238.05 (38.88)	0.468	
bcarot (mean (SD))		260.81 (130.38)	191.78 (94.42)	224.00 (87.56)	232.93 (112.54)	217.81 (122.34)	0.767	

Table 1: Summary of Data by Dose Group

4.2 Association between dosage of beta carotene and serum beta carotene levels

Our two-stage analysis revealed clear evidence that beta-carotene supplementation significantly increases serum beta-carotene levels in a dose-dependent manner. First, comparing supplemented subjects (15-60 dose groups) versus placebo (0 dose group), we found a highly significant increase in beta-carotene levels during the treatment period (mean difference: 645-1300 BCL, 95% CI: 645-1300; $p = 6.943 \times 10^{-5}$). This robust effect was confirmed through Welch's t-test, which accounted for potential variance differences between groups. Building on this, our dose-specific analysis demonstrated that the magnitude of increase varied substantially across dosage levels (ANOVA p-value = 0.0483). Post-hoc comparisons indicated particularly strong effects at higher doses (45 and 60 dose), which consistently outperformed lower doses (15 and 30 dose) in elevating beta-carotene concentrations. These results collectively demonstrate that while any supplementation produces measurable increases in serum beta-carotene, the effect is not uniform across doses - higher doses elicit substantially greater responses. The dose-response relationship suggests potential clinical relevance for precision dosing strategies, though we note that our limited subgroup sample sizes may limit the precision of between-dose comparisons.

4.3 Relationship between dosage of beta carotene and beta carotene levels over time

Term	Estimate	Std. Error	df	t-value	p-value
(Intercept)	1361.912	786.835	47.671	1.731	0.090
treatmentNot Dose 0	636.682	170.766	237.796	3.728	0.000
month	-15.836	20.749	221.142	-0.763	0.446
vite	25.975	21.993	259.966	1.181	0.239

age	-6.756	9.663	45.510	-0.699	0.488
maleMale	-174.529	76.761	45.300	-2.274	0.028
bmi	-29.576	13.460	44.872	-2.197	0.033
chol	0.743	1.237	50.052	0.601	0.551
treatmentNot Dose 0:month	36.297	22.750	218.761	1.595	0.112

Table 2: Summary Statistics of Placebo vs Supplementation

Our analysis revealed that beta-carotene supplementation significantly impacts serum beta-carotene levels initially. Comparing subjects receiving supplementation (non-zero dose) to the placebo group, we observed a substantial increase in serum beta-carotene levels (mean difference: 636.66, p-value < 0.001). However, the interaction between treatment and time (month) was not statistically significant with a p-value of 0.112, suggesting that the effect of supplementation over time does not differ between subjects that have a supplement of beta carotene and subjects that have the placebo which can be evidenced by figure [A].

Term	Estimate	Std. Error	df	t-value	p-value
(Intercept)	891.771	690.272	38.555	1.292	0.204
dose30	194.499	198.236	207.470	0.981	0.328
dose45	238.947	204.281	208.009	1.170	0.243
dose60	554.603	198.256	207.456	2.797	0.006
month	35.141	19.544	180.320	1.798	0.074
age	4.463	8.519	36.380	0.524	0.604

maleMale	-157.384	69.834	36.461	-2.254	0.030
bmi	-18.210	12.597	36.070	-1.446	0.157
chol	1.587	1.054	36.217	1.506	0.141
dose30:month	-8.396	27.492	177.426	-0.305	0.760
dose45:month	-20.063	28.342	177.295	-0.708	0.480
dose60:month	-43.990	27.492	177.426	-1.600	0.111

Table 3: Summary Statistics of All Dose Supplementations

Our analysis revealed no evidence of dose-dependent effects. The interaction between each of the doses and time (month) revealed nuanced trends in serum beta-carotene levels. While the 60 dose group showed a significant initial increase in beta-carotene levels shown with a p-value of 0.006, its interaction with time suggests similar patterns of beta carotene levels over time with dose 15, as do doses 30 and 45 compared to dose 15. Though this monthly decrease did not reach statistical significance, the magnitude of the effect in terms of dose 60 with time with an estimate of -44 implies a potential difference of the treatment effect over time at higher doses. For the 45 dose and 30 dose groups, neither the baseline dose effect nor their interactions with time were significant with p-values above 0.2 for all estimates. However, the 45 dose group exhibited a numerically larger monthly decline of -20 compared to the 30 dose group of -8.4 when both are compared to baseline dose 15, hinting at possible dose-specific decay patterns. The lack of significance for these

trends may reflect limited statistical power or true null effects with the lack of unique or significant trend from one dose compared to the others shown in figure [B].

4.4 Relationship between dosage of beta carotene and rate back to baseline

Term	Estimate	Std. Error	p-value				
Intercept	1892.746	1184.027	0.118	BMI	-37.290	22.796	0.110
Dose 30	72.444	255.345	0.778	Cholesterol (mg/dL)	-0.447	1.769	0.802
Dose 45	160.435	270.674	0.556	Dose30 x Time interaction	62.538	125.257	0.618
Dose 60	424.760	254.838	0.102	Dose45 x Time interaction	274.559	131.169	0.038
month_centered	-44.671	89.065	0.617	Dose60 x Time interaction	106.034	122.888	0.390
Spline Month 12	-281.671	130.364	0.032	Dose30 x Spline interaction	-128.394	185.723	0.490
Age (years)	9.036	14.241	0.530	Dose45 x Spline interaction	-497.902	192.971	0.011
Male	-181.823	122.772	0.147	Dose60 x Spline interaction	-303.729	180.487	0.094

Table 4: Summary Statistics of Months 10-15 for LLM model

Our analysis and figure [C] revealed significant dose-dependent differences in how serum beta-carotene levels returned to baseline after supplementation was discontinued. The 45 dose group demonstrated the most rapid decline, with an approximate 222 reduction in beta carotene level per month at 12 months in the post supplementation phase more than dose 15 with p-value of 0.011, suggesting accelerated decrease at this dose. In contrast, the 60 dose group showed a slower, but almost significant trend toward baseline with a p-value of 0.094 and an estimate of 197 beta carotene level decrease more per month post month 12 than dose 15. The lack of significant decline in the 30 dose group compared to dose 15 group could reflect adjustments at lower doses.

4.5 Relationship of age, male, bmi, and cholesterol with interaction of month and dose

Model	term	npar	AIC	BIC	logLik	X.2.log.L.	statistic	df	p.value
Age Interaction	reduced_model1	16	7179.364	7246.474	-3573.682	7147.364	NA	NA	NA
Age Interaction	age_model	23	7190.171	7286.643	-3572.086	7144.171	3.192592	7	0.8666327
Male Interaction	reduced_model2	16	7179.364	7246.474	-3573.682	7147.364	NA	NA	NA
Male Interaction	male_model	23	7170.939	7267.411	-3562.470	7124.939	22.424705	7	0.0021455
BMI Interaction	reduced_model3	14	7182.572	7241.294	-3577.286	7154.572	NA	NA	NA
BMI Interaction	bmi_model	23	7186.495	7282.966	-3570.247	7140.495	14.077071	9	0.1196100
Chol Interaction	reduced_model4	16	7179.364	7246.474	-3573.682	7147.364	NA	NA	NA
Chol Interaction	chol_model	23	7189.137	7285.608	-3571.568	7143.137	4.226950	7	0.7532976

Table 5: Results of 3 way anova for Baseline Covariates

The likelihood ratio tests revealed that gender significantly modified the effect of beta-carotene supplementation over time ($\chi^2=22.42$, p-value = 0.002), indicating that males and females experienced different patterns of serum beta-carotene level changes during treatment over time. In contrast, age ($\chi^2=3.19$, p-value=0.867), BMI ($\chi^2=14.08$, p-value=0.120), and cholesterol ($\chi^2=4.23$, p-value=0.753) did not show statistically significant interactions with supplementation effects over time. The model fit statistics (AIC/BIC) consistently favored the reduced models for age, BMI

and cholesterol, further supporting the absence of meaningful effect modification by these factors. These results suggest that while gender plays an important role in shaping the response to beta-carotene supplementation, the treatment effects appear consistent across different age groups, BMI categories, and cholesterol levels. Visuals of the differences for each individual baseline covariate can be found in figure [D] for age, figure [E] for male, figure [F] for BMI, and figure [G] for Chol.

4.6 Beta carotene level Prediction Modeling for months 10-12

id	dose	month_centered	pred	lower_ci	upper_ci
11	15 Dose Group	10	1356.651	806.8288	1935.857
11	15 Dose Group	11	1443.191	830.3806	1994.382
11	15 Dose Group	12	1496.936	827.1131	2093.816
48	15 Dose Group	10	1281.716	672.2049	1820.118
48	15 Dose Group	11	1306.056	753.3280	1941.010
48	15 Dose Group	12	1364.730	720.1930	2037.910
3	30 Dose Group	10	1235.744	608.2324	1814.971
3	30 Dose Group	11	1258.478	661.2456	1897.877
3	30 Dose Group	12	1297.550	688.4717	1916.237
27	30 Dose Group	10	1514.188	925.8470	2085.064
27	30 Dose Group	11	1491.141	889.6896	2073.567
27	30 Dose Group	12	1485.237	893.3262	2136.330
12	45 Dose Group	10	1372.736	772.4489	1992.430
12	45 Dose Group	11	1369.825	741.6881	1972.817
12	45 Dose Group	12	1402.564	730.0137	2047.254
23	45 Dose Group	10	1528.966	899.3745	2119.642
23	45 Dose Group	11	1536.246	942.4472	2165.159
23	45 Dose Group	12	1542.447	878.5688	2194.315
20	60 Dose Group	10	1433.459	879.8311	2001.531
20	60 Dose Group	11	1435.921	873.6876	2096.133
20	60 Dose Group	12	1438.205	805.5143	2081.240
36	60 Dose Group	10	1318.797	758.1745	1881.963
36	60 Dose Group	11	1314.327	720.2095	1927.639
36	60 Dose Group	12	1278.730	650.5119	1927.290

Table 6: Prediction Estimates Month 10-12 for 2 randomly selected subjects per Dose

The predicted serum beta-carotene levels for two randomly selected patients per dose group showed distinct patterns during the extended 3-month treatment period. In the 15 dose group, subject 11 exhibited a steady increase from 1,357 to 1,497 beta carotene level (BCL), while subject 48 had a more modest rise from 1,282 to 1,365 BCL, with both showing wide confidence intervals (± 500 -600 BCL). The 30 dose group demonstrated relative stability, with Patient 3 increasing slightly from 1,236 to 1,298 BCL and subject 27 maintaining consistent levels around 1,485-1,514 BCL. For the 45 dose group, subject 12 stayed near constant at approximately 1,400 BCL, whereas subject 23 sustained higher levels ($\sim 1,530$ -1,542 BCL). The 60 dose group revealed the most stable predictions with subject 20's levels remaining at around 1,435–1,438 BCL, while subject 36 showed a slight decline to 1,279 BCL, though both had overlapping uncertainty ranges (± 600 -650 BCL). Across all doses, higher concentrations (45 and 60 doses) generally produced more sustained levels, but substantial inter-patient variability (shown by the wide confidence intervals) suggesting individual factors may critically influence treatment outcomes and can vary depending on the subject(s) chosen for the prediction model with their linear trends shown in figure [H].

5 Discussion

The analyses reveal a complex dose-response relationship for beta-carotene supplementation, with the 60 dose showing the strongest initial effect but potential diminishing returns over time, while the 45 dose demonstrated a sharp decline after supplementation. Lower doses (15 and 30 doses) showed minimal effects. Notably, gender significantly modified treatment interaction with time, though age, BMI and cholesterol did not appear influential. The substantial inter-subject variability in predicted levels during extended treatment, evidenced by wide confidence intervals across all dose groups, suggests individual factors may outweigh dose effects in determining outcomes. Several limitations temper these findings, including unmeasured confounding variables and model assumptions that may not fully capture the complex patterns. Future research should hold longer-term clinical trials to validate observed patterns and increase sample size to reduce variation in results. These findings collectively highlight that while beta-carotene supplementation clearly elevates serum levels, the response is neither uniform across doses nor consistent over time, underscoring the need for more nuanced approaches to supplementation protocols.

6 Appendix

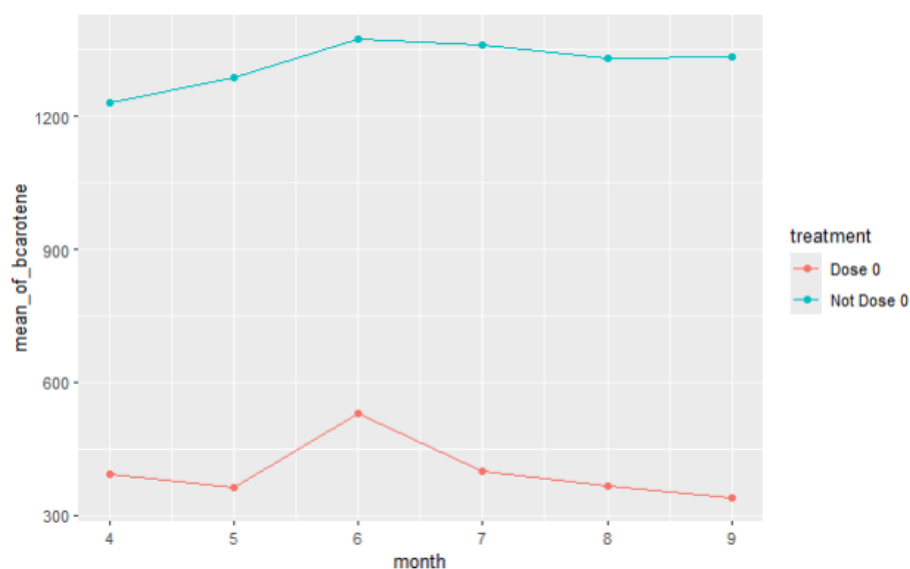


Figure A: Linear Mixed model plot of mean beta carotene levels for subjects that have a supplementation of beta carotene and subjects that have placebo over the course of months 4-9

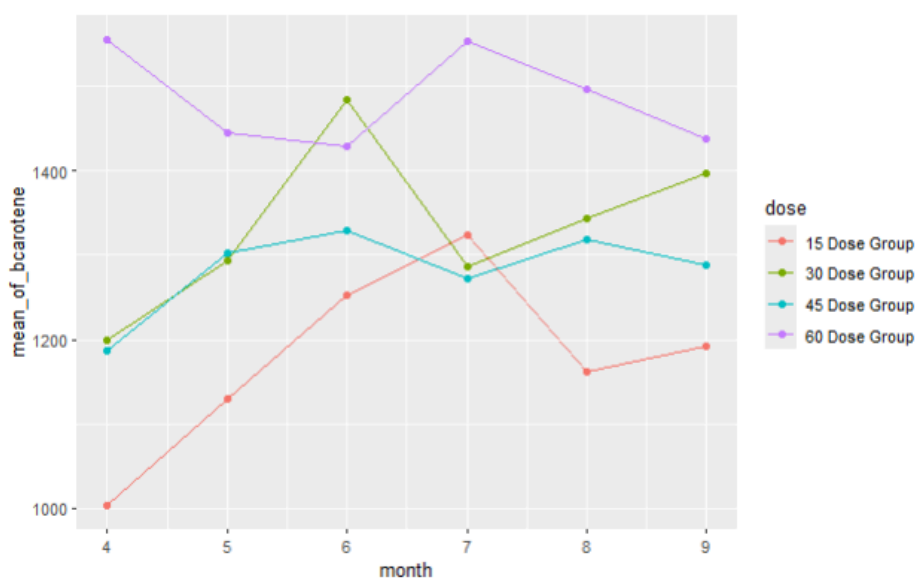


Figure B: Linear Mixed Model plot of mean beta carotene levels for the different doses of beta carotene supplementation over the course of months 4-9

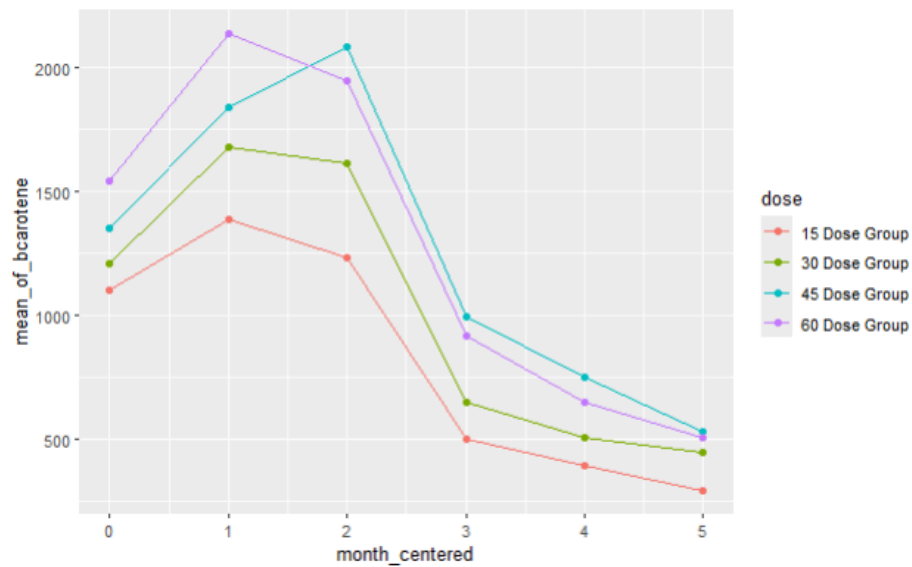


Figure C: Linear Mixed Model plot of mean beta carotene levels for the different doses of beta carotene supplementation over the course of months 10-15 (centered for month 10 to be baseline here)

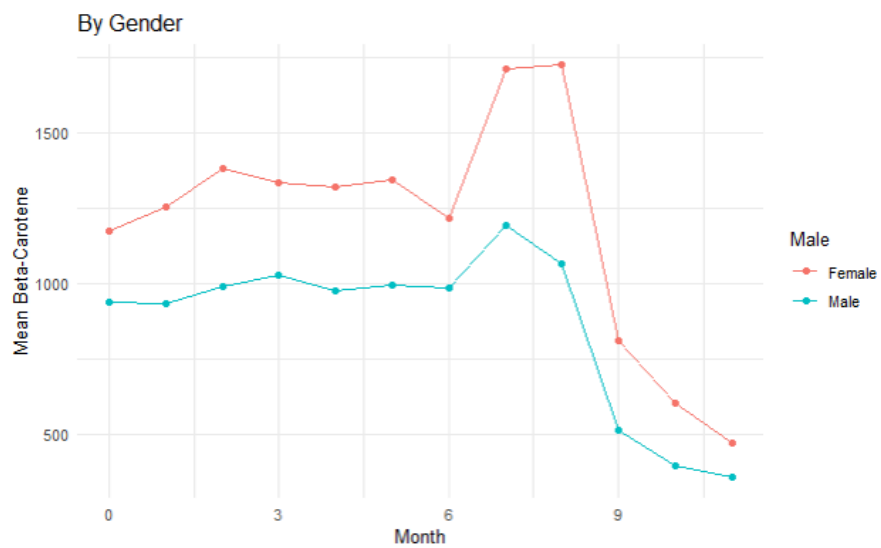


Figure D: Linear Mixed Model plot of mean beta carotene levels for each gender over the course of months 0-15

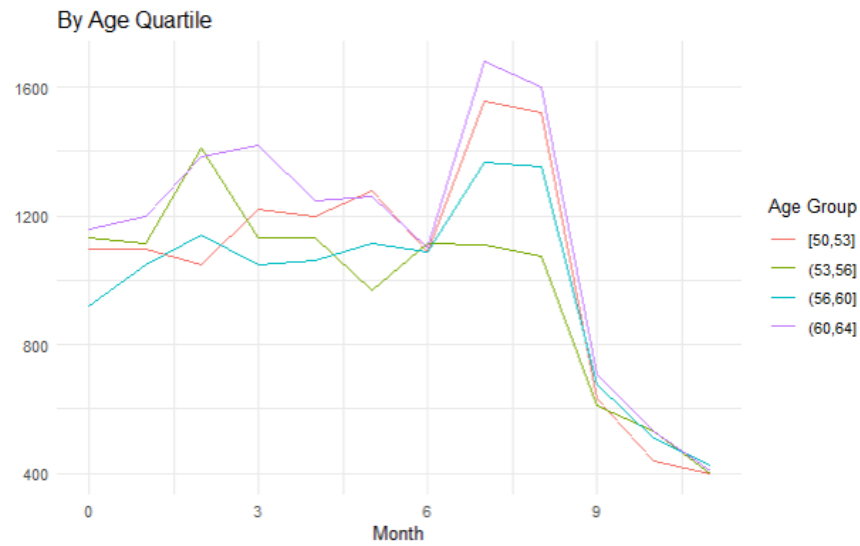


Figure E: Linear Mixed Model plot of mean beta carotene levels for age split into quartiles over the course of months 0-15

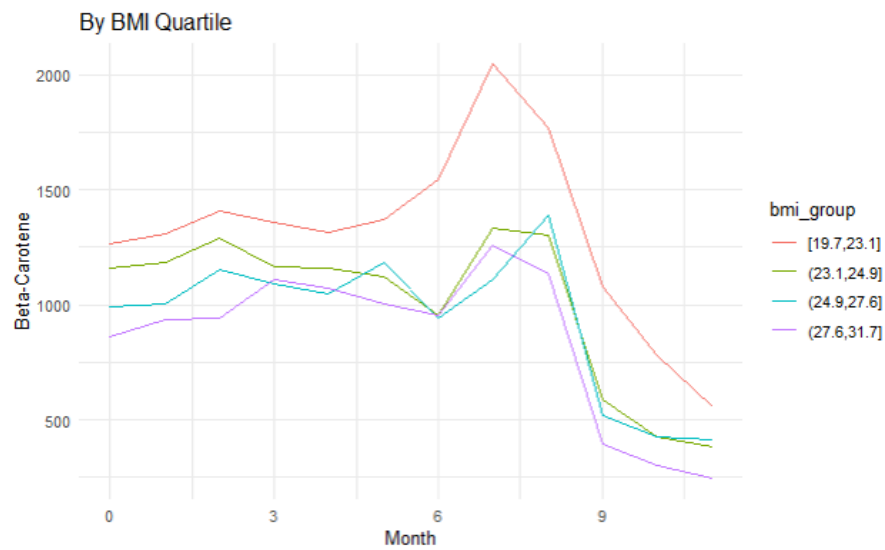


Figure F: Linear Mixed Model plot of mean beta carotene levels for BMI split into quartiles over the course of months 0-15

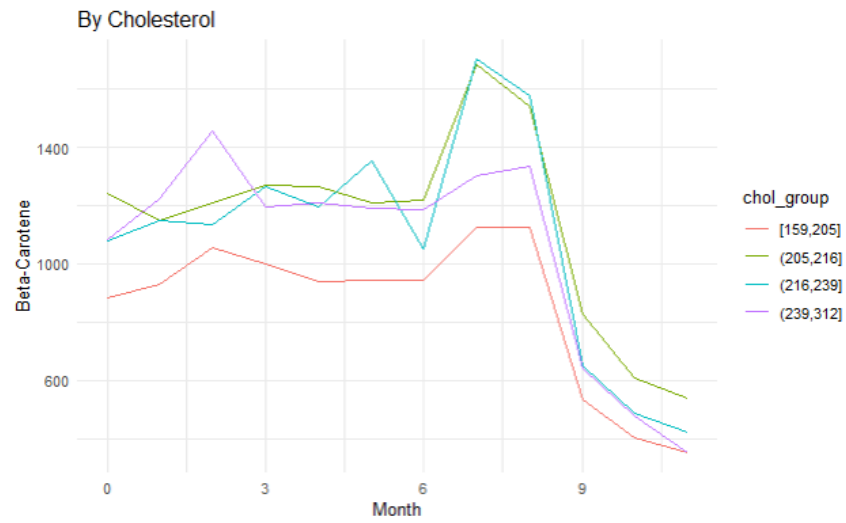


Figure G: Linear Mixed Model plot of mean beta carotene levels for Chol split into quantiles over the course of months 0-15

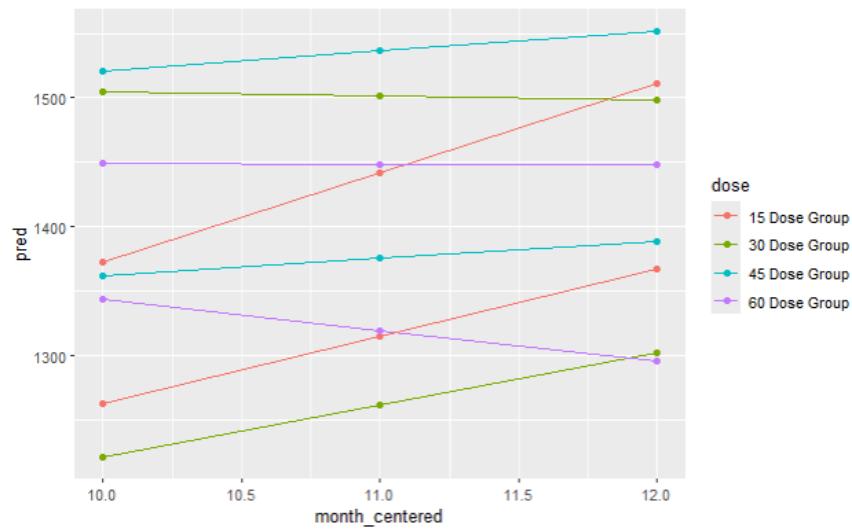


Figure H: Prediction model of beta carotene levels for 2 randomly selected subjects per dose group for months 10-12 given months 4-9

References

- [1] Grune, Tilman, et al. "Beta-Carotene Is an Important Vitamin A Source for Humans." *The Journal of Nutrition*, U.S. National Library of Medicine, 27 Oct. 2010, [pmc.ncbi.nlm.nih.gov/articles/PMC3139236/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC3139236/).
- [2] Open Resources for Nursing (Open RN). "Chapter 1 Pharmacokinetics & Pharmacodynamics." *Nursing Pharmacology* [Internet]. 2nd Edition., U.S. National Library of Medicine, 1 Jan. 1970, www.ncbi.nlm.nih.gov/books/NBK595006/.
- [3] Grogan, Sean. "Pharmacokinetics." *StatPearls [Internet]*., U.S. National Library of Medicine, 30 July 2023, www.ncbi.nlm.nih.gov/books/NBK557744/.
- [4] "Atrain Education." 5. Pharmacokinetics and the Aging Process | ATrain Education, www.etrainceu.com/content/5-pharmacokinetics-and-aging-process. Accessed 11 June 2025.